Safety and Connectivity Certification for Finite-Time Swarm Formation Tracking

Yash Purswani, Shivansh Agarwal and Anuj Tiwari

***Abstract*— Finite-time formation tracking is useful for coordinating multi-agent systems in time-critical missions, but convergence to a prescribed formation does not guarantee collision avoidance or communication connectivity during transients. Existing approaches address finite-time coordination and safety-constrained formation control, yet systematic connections between transient disagreement, formation geometry, and coupling-gain selection remain important for certifying safe maneuvers. This paper develops a disagreement-based framework for certifying transient safety and connectivity during finite-time tracking of arbitrary prescribed formation trajectories. Local feedback ensures exact finite-time tracking with a graph-independent deadline, while cooperative coupling accelerates disagreement contraction. The resulting graph-dependent bound provides sufficient geometric conditions for collision avoidance and connectivity preservation, together with an offline coupling-gain selection rule based on the available clearance. Numerical simulations demonstrate improved transient convergence with increased coupling, the associated control-effort trade-off, and the effectiveness and conservatism of the proposed certificates.**

***Index Terms*— Multi-agent systems, formation control, finite-time stability, collision avoidance, connectivity preservation.**

I. Introduction

Multi-agent systems enable tasks such as cooperative transportation, autonomous navigation, and distributed sensing, where multiple agents must follow a moving reference while maintaining a prescribed spatial arrangement [1]–[3]. Formation-control frameworks commonly exploit inter-agent interactions and graph-based coordination to regulate collective geometry [4], [5]. In practical missions, however, formations may be asymmetric or time-varying, requiring each agent to track an individually prescribed formation position as the formation translates, rotates, or deforms.

For time-critical maneuvers, asymptotic convergence may be insufficient, motivating finite-time tracking with an explicit settling deadline [6], [7]. However, finite-time convergence alone does not ensure safe transient behaviour: agents may collide or lose communication links before reaching their prescribed formation positions. Cooperative interactions can reduce relative tracking errors, but their contribution to transient safety must be quantified. This motivates relating disagreement contraction to collision avoidance and connectivity preservation throughout the maneuver.

Finite-time consensus and formation control have established convergence guarantees under varied interaction structures [8], [9]. Fedele and D'Alfonso [10] studied finite-time centroid tracking with bounded aggregation, while collision avoidance and connectivity preservation have been addressed through energy-based, optimization, and barrier-function approaches [11]–[14]. In contrast, this work investigates how cooperative disagreement contraction can provide explicit transient safety and connectivity certificates for individually prescribed formation trajectories, without modifying the nominal tracking protocol or requiring online safety optimization.

This paper considers a distributed first-order kinematic protocol for tracking arbitrary prescribed C^1 formation trajectories. Local feedback guarantees exact finite-time formation-position tracking with a graph-independent deadline, while cooperative coupling accelerates disagreement decay without altering this guarantee. A graph-dependent analysis bounds relative tracking errors and translates them into sufficient geometric conditions for pairwise collision avoidance and preservation of a designer-selected connected spanning subgraph. These conditions further provide an offline coupling-gain selection rule based on the planned formation geometry and initial disagreement. The protocol generates velocity setpoints for an underlying low-level controller or autopilot.

The main contributions are:

1) A finite-time formation-position tracking framework that preserves an explicit graph-independent settling bound under cooperative coupling for arbitrary prescribed C^1 trajectories.
2) A graph-dependent exponential disagreement bound for undirected time-varying communication graphs, quantifying the transient contraction induced by cooperative coupling.
3) Sufficient geometric conditions linking disagreement bounds to pairwise collision avoidance and preservation of a connected spanning subgraph throughout the maneuver.
4) An offline coupling-gain selection rule based on the time-varying geometric clearance and initial disagreement, demonstrated through numerical examples highlighting certificate effectiveness and conservatism.

II. Kinematic Model

Consider $n \geq 2$ agents with positions $x_i \in \mathbb{R}^d$. The desired formation position of agent i is

$$x_{d,i}(t) = r(t) + p_i(t), \qquad \delta_i(t) = x_i(t) - x_{d,i}(t), \quad (1)$$

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where $r(t) \in \mathbb{R}^d$ is the reference trajectory and $p_i(t) \in \mathbb{R}^d$ is the desired offset. The prescribed formation trajectories are assumed C^1 with locally bounded derivatives, allowing the formation to translate and deform over time.

The distributed kinematic protocol is

$$\dot{x}_i = -\delta_i - \Phi_\beta(\delta_i) - \sum_{j=1}^{n} w_{ij} M(\delta_i - \delta_j) + \dot{x}_{d,i}, \quad (2)$$

where $\dot{x}_i$ is the ideal commanded velocity, $w_{ij} \geq 0$ are interaction weights, and $M \in \mathbb{R}^{d\times d}$ is symmetric positive semidefinite. For $0 < \beta < 1$, the component-wise fractional-power feedback is

$$[\Phi_\beta(z)]_k = \operatorname{sign}(z_k)|z_k|^\beta, \qquad k = 1, \ldots, d, \quad (3)$$

with $\Phi_\beta(0) = 0$.

Interactions are described by a time-varying undirected graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$, where $\mathcal{V} = \{1, \ldots, n\}$ and $\mathcal{E}(t)$ denotes the active communication links [15], [16]. As agents move, links may be established or lost when inter-agent distances cross the communication range, changing the graph topology while agent positions remain continuous. With symmetric weights $w_{ij} = w_{ji} \geq 0$ and $w_{ii} = 0$, the weighted Laplacian is

$$L_w = \operatorname{diag}\left(\sum_{j=1}^{n} w_{ij}\right) - W, \qquad W = [w_{ij}]. \quad (4)$$

The range-dependent weights are

$$w_{ij}(x) = \frac{a_{ij}(x)}{\|x_i - x_j\|^\nu + \varepsilon_w}, \qquad \nu \geq 1, \quad \varepsilon_w > 0, \quad (5)$$

where

$$a_{ij}(x) = \begin{cases} 1, & \|x_i - x_j\| \leq R_c, \\ 0, & \|x_i - x_j\| > R_c, \end{cases} \quad a_{ij} = a_{ji}, \quad a_{ii} = 0. \quad (6)$$

Here R_c is the communication radius and ε_w prevents singular weights. A connected spanning subgraph $\mathcal{H}$ is selected for preservation.

From (1), the relative errors satisfy

$$\delta_i - \delta_j = (x_i - x_j) - (x_{d,i} - x_{d,j}), \quad (7)$$

allowing the coupling to be computed from relative positions and formation position assignments.

Stacking $\delta = [\delta_1^\top, \ldots, \delta_n^\top]^\top$, the error dynamics become

$$\dot{\delta} = -\delta - \Phi_\beta(\delta) - (L_w(x) \otimes M)\delta, \quad (8)$$

where $\Phi_\beta(\delta)$ denotes the stacked fractional-power feedback.

The discontinuous dynamics are interpreted in the Filippov sense, $\dot{\delta} \in \mathcal{F}[f](t, \delta)$, where f is the right-hand side of (8) and $\mathcal{F}[f](t, \delta) = \bigcap_{r>0} \bigcap_{N:|N|=0} \overline{\text{co}}\{f(t, z) : z \in B(\delta, r) \setminus N\}$. Bounded weights make f measurable and locally essentially bounded; its growth is at most linear. Thus, solutions exist from every finite initial condition, [17]. Along a Filippov solution, the coupling can be written with an effective Laplacian $\widetilde{L}_w$, given by the convex combination of limiting weighted Laplacians. Hence $\widetilde{L}_w^T = \widetilde{L}_w \succeq 0$ and $\widetilde{L}_w \mathbf{1} = 0$.

Assumption 1. *Each agent knows its own state and formation position trajectory, including $\dot{x}_{d,i}$, and exchanges state and formation position information with active neighbours. Thus, δ_i and $\delta_i - \delta_j$ are locally computable.*

Problem statement: Develop a distributed kinematic protocol that (i) achieves exact finite-time formation position tracking with a graph-independent deadline; (ii) provides sufficient conditions for collision avoidance and preservation of the connected spanning subgraph $\mathcal{H}$; and (iii) enables offline coupling-gain selection from the planned desired formation geometry and initial disagreement.

III. Finite-Time Formation-Position and Centroid Tracking

Let $\delta_0 = \delta(t_0)$ and define the non-negative envelope

$$\mathcal{R}(t) = \left[\left(1 + \|\delta_0\|^{1-\beta}\right) e^{-(1-\beta)(t-t_0)} - 1\right]_+^{1/(1-\beta)}, \quad (9)$$

where $[a]_+ = \max\{a, 0\}$. The finite-time arrival deadline is

$$T = t_0 + \frac{\ln(1 + \|\delta_0\|^{1-\beta})}{1-\beta}. \quad (10)$$

Theorem 1 (Finite-time formation-position tracking)**.** *Under Assumption 1, with $M = M^\mathsf{T} \succeq 0$ and $0 < \beta < 1$, every Filippov solution of* (8) *satisfies*

$$\|\delta(t)\| \leq \mathcal{R}(t), \qquad t \geq t_0 \quad (11)$$

and $\delta(t) = 0$ for all $t \geq T$. Thus, $x_i(t) = x_{d,i}(t)$ for all agents and all $t \geq T$. The deadline is independent of the interaction graph, its connectivity, and M.

Proof. Consider $V = \frac{1}{2}\|\delta\|^2$. For almost every time, symmetry and positive semi-definiteness of the effective Laplacian and M give

$$\begin{aligned} \mathcal{D} &:= \delta^\mathsf{T}(\widetilde{L}_w \otimes M)\delta \\ &= \frac{1}{2}\sum_{i,j} \widetilde{w}_{ij}(\delta_i - \delta_j)^\mathsf{T} M(\delta_i - \delta_j) \geq 0. \end{aligned} \quad (12)$$

Since $1+\beta < 2$, norm monotonicity implies $\sum_{\ell=1}^{nd} |\delta_\ell|^{1+\beta} \geq \|\delta\|_2^{1+\beta}$. Consequently along (8),

$$\dot{V} = -\|\delta\|^2 - \delta^\mathsf{T}\Phi_\beta(\delta) - \delta^\mathsf{T}(\widetilde{L}_w \otimes M)\delta \quad (13)$$

$$\leq -\|\delta\|^2 - \|\delta\|^{1+\beta} = -2V - (2V)^{(1+\beta)/2}, \quad (14)$$

Let $z = \|\delta\| = \sqrt{2V}$. Whenever $z > 0$, (14) implies $\dot{z} \leq -z - z^\beta$. With $y = z^{1-\beta}$,

$$\dot{y} \leq -(1-\beta)(y+1). \quad (15)$$

Integration yields $y(t) + 1 \leq (y(t_0) + 1)e^{-(1-\beta)(t-t_0)}$ until the tracking error reaches zero. This comparison bound establishes the bounds in (9) and (10), guaranteeing that the error reaches zero within a finite time. Since the origin is invariant under the error dynamics, the error remains zero after reaching it. Since V is non-increasing, the error remains bounded. □

A. Centroid and Formation Guarantees

Assume the formation position offsets are centered:

$$\sum_{i=1}^{n} p_i(t) = 0, \qquad \forall t \geq t_0. \tag{16}$$

Let $c(t) = \frac{1}{n}\sum_i x_i(t)$ be the swarm centroid. Since $x_{d,i} = r + p_i$, the mean desired formation position is r.

Corollary 1 (Reference tracking)**.** *Under the centering condition* (16)*,*

$$\|c(t) - r(t)\| \leq \frac{\mathcal{R}(t)}{\sqrt{n}}, \qquad t \geq t_0. \tag{17}$$

Consequently, $c(t) = r(t)$ *for all* $t \geq T$.

Proof. Since $x_{d,i} = r + p_i$ and $\sum_{i=1}^{n} p_i = 0$, the centroid error satisfies

$$c - r = \frac{1}{n}\sum_{i=1}^{n}(x_i - x_{d,i}) = \frac{1}{n}\sum_{i=1}^{n}\delta_i.$$

By the Cauchy–Schwarz inequality,

$$\|c - r\| \leq \frac{1}{\sqrt{n}}\left(\sum_{i=1}^{n}\|\delta_i\|^2\right)^{1/2} = \frac{\|\delta\|}{\sqrt{n}}.$$

Applying the bound $\|\delta(t)\| \leq \mathcal{R}(t)$ from Theorem 1 gives the stated inequality. Since $\mathcal{R}(t) = 0$ for all $t \geq T$, it follows that $c(t) = r(t)$ for all $t \geq T$. □

Remark 1. *For uncentered offsets, the centroid converges to their mean desired position,* $r + \frac{1}{n}\sum_i p_i$.

IV. Disagreement-Based Safety and Connectivity Certificates

The complete formation position error contains both a common component and a disagreement component. Only disagreement changes relative positions. Define

$$\begin{aligned} &\Pi = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad \eta = (\Pi \otimes I_d)\delta, \\ &\mu = \lambda_{\min}(M) \geq 0. \end{aligned} \tag{18}$$

Thus $\eta_i = \delta_i - \bar{\delta}$ and $\delta_i - \delta_j = \eta_i - \eta_j$.

Theorem 2 (Graph-dependent disagreement contraction)**.** *For every Filippov solution at* $t \in [t_0, T]$*, the disagreement satisfies*

$$\|\eta(t)\| \leq \|\eta(t_0)\| \exp\left[-\int_{t_0}^{t}\left(1 + \mu\lambda_2(\widetilde{L}_w(s))\right) ds\right]. \tag{19}$$

Moreover, $\eta(t) = 0$ *for all* $t \geq T$*. The bound remains valid across communication topology changes.*

Proof. Since $\Pi\widetilde{L}_w = \widetilde{L}_w$ and $\Pi\eta = \eta$, the disagreement dynamics follow from (8) as

$$\dot{\eta} = -\eta - (\Pi \otimes I_d)\Phi_\beta(\delta) - (\widetilde{L}_w \otimes M)\eta. \tag{20}$$

For $W_\eta = \frac{1}{2}\|\eta\|^2$, using $\eta^{\mathsf{T}}(\Pi \otimes I_d) = \eta^{\mathsf{T}}$ yields

$$\dot{W}_\eta = -\|\eta\|^2 - \eta^{\mathsf{T}}\Phi_\beta(\delta) - \eta^{\mathsf{T}}(\widetilde{L}_w \otimes M)\eta. \tag{21}$$

By component-wise monotonicity of Φ_β,

$$\eta^{\mathsf{T}}\Phi_\beta(\delta) = \frac{1}{2n}\sum_{i,j}(\delta_i - \delta_j)^{\top}\big(\Phi_\beta(\delta_i) - \Phi_\beta(\delta_j)\big) \geq 0. \tag{22}$$

Since $M \succeq 0$ and $\mu = \lambda_{\min}(M)$, the Rayleigh quotient bound gives

$$\eta^{\top}(\widetilde{L}_w \otimes M)\eta \geq \mu\lambda_2(\widetilde{L}_w)\|\eta\|^2.$$

Consequently,

$$\dot{W}_\eta \leq -2\big[1 + \mu\lambda_2(\widetilde{L}_w)\big]W_\eta.$$

Applying the comparison lemma and integrating yields (19). Finally, Π is an orthogonal projection, so $\|\eta(t)\| \leq \|\delta(t)\|$. By Theorem 1, $\delta(t) = 0$ for all $t \geq T$, which implies $\eta(t) = 0$ for all $t \geq T$. □

A. Pairwise Distance Bounds

Let $d_{ij}^q(t) = \|x_{d,i}(t) - x_{d,j}(t)\|$ and $d_{ij}^x(t) = \|x_i(t) - x_j(t)\|$.

Lemma 1 (Relative-distance envelope)**.** *For every pair* i, j*,*

$$d_{ij}^q(t) - \sqrt{2}\|\eta(t)\| \leq d_{ij}^x(t) \leq d_{ij}^q(t) + \sqrt{2}\|\eta(t)\|. \tag{23}$$

Proof. Since $x_i - x_j = (x_{d,i} - x_{d,j}) + (\eta_i - \eta_j)$, the triangle inequality gives

$$\|x_i - x_j\| \leq \|x_{d,i} - x_{d,j}\| + \|\eta_i - \eta_j\|,$$

while the reverse-triangle inequality yields

$$\|x_i - x_j\| \geq \|x_{d,i} - x_{d,j}\| - \|\eta_i - \eta_j\|.$$

Moreover,

$$\|\eta_i - \eta_j\| \leq \|\eta_i\| + \|\eta_j\| \leq \sqrt{2(\|\eta_i\|^2 + \|\eta_j\|^2)} \leq \sqrt{2}\|\eta\|.$$

Hence, the two inequalities above establish the claimed bounds. □

A connected spanning subgraph $\mathcal{H} = (\mathcal{V}, \mathcal{E}_\mathcal{H})$ is specified by the designer to preserve communication links essential for coordination during formation maneuvers. Let $L_\mathcal{H}$ denote its unweighted Laplacian. The formation position-edge length envelope and the minimum pairwise formation position distance are defined as

$$\begin{aligned} \ell_\mathcal{H}(t) &= \max_{(i,j)\in\mathcal{E}_\mathcal{H}} \|x_{d,i}(t) - x_{d,j}(t)\|, \\ d_q(t) &= \min_{i<j} \|x_{d,i}(t) - x_{d,j}(t)\|. \end{aligned} \tag{24}$$

Let $\lambda_2(L_\mathcal{H}) > 0$ and let $\underline{w} > 0$ be a lower bound on every active selected subgraph-edge weight. For the range-dependent weights in (5), one may choose $\underline{w} = (R_c^\nu + \varepsilon_w)^{-1}$. Define

$$\begin{aligned} \gamma_\mathcal{H} &= 1 + \mu\underline{w}\lambda_2(L_\mathcal{H}), \\ \rho_0 &= \sqrt{2}\|\eta(t_0)\|. \end{aligned} \tag{25}$$

Since $\eta = (\Pi \otimes I_d)\delta$ and Π is an orthogonal projection, it cannot increase the Euclidean norm; hence $\|\eta(t)\| \leq \|\delta(t)\|$. By Theorem 1, this gives $\sqrt{2}\|\eta(t)\| \leq \sqrt{2}\mathcal{R}(t)$. Moreover, Theorem 2 gives exponential contraction of the

disagreement. While all selected subgraph edges remain active, $\widetilde{L}_w \succeq \underline{w} L_{\mathcal{H}}$, so the contraction rate is at least $\gamma_{\mathcal{H}}$. Consequently,

$$\rho_{\mathcal{H}}(t) = \min\left\{\sqrt{2}\,\mathcal{R}(t),\, \rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)}\right\}. \tag{26}$$

This is an a priori envelope on the maximum pairwise disagreement for every pair of agents, provided the selected subgraph remains available.

Theorem 3 (Joint collision avoidance and connectivity preservation)**.** *If the planned formation positions satisfy, for all $t \geq t_0$,*

$$d_q(t) > d_{\text{safe}} + \rho_{\mathcal{H}}(t), \tag{27}$$

$$\ell_{\mathcal{H}}(t) < R_c - \rho_{\mathcal{H}}(t), \tag{28}$$

then every pair of agents remains strictly separated by more than d_{safe}, every selected subgraph edge remains within range R_c, and the communication graph remains connected. The formation position-tracking deadline T in (10) *is unchanged.*

Proof. At t_0, (26) gives $\rho_{\mathcal{H}}(t_0) = \rho_0$, since $\|\eta(t_0)\| \leq \|\delta(t_0)\|$. By Lemma 1 and (27), all actual pairwise distances exceed d_{safe} initially. The same lemma and (28) put all selected subgraph links strictly inside R_c initially.

Suppose a selected subgraph edge first reaches R_c at time $t_e > t_0$. Until t_e, all selected subgraph edges remain active, so the lower bound $\widetilde{L}_w \succeq \underline{w} L_{\mathcal{H}}$ and Theorem 2 imply (26). By continuity, the bound also holds at t_e. For every selected subgraph edge, Lemma 1 then gives $\|x_i(t_e) - x_j(t_e)\| \leq \ell_{\mathcal{H}}(t_e) + \rho_{\mathcal{H}}(t_e) < R_c$, a contradiction. Thus the selected subgraph persists for all time. Finally, the lower half of Lemma 1 and (27) imply $d_{ij}^x(t) > d_{\text{safe}}$ for every pair and all times. The formation position-tracking theorem is unaffected by the graph argument. □

B. Coupling-Gain Design Rule

Define the available geometric margin

$$s(t) = \min\{d_q(t) - d_{\text{safe}},\, R_c - \ell_{\mathcal{H}}(t)\}. \tag{29}$$

The margin depends only on the prescribed formation position trajectories, the selected safety threshold, the operational communication range, and the selected subgraph. It can therefore be computed offline, before execution, without live agent-state measurements. Geometric feasibility requires $s(t) > 0$ for all $t \geq t_0$. Suppose $0 < \rho_0 < s(t_0)$. Since exact formation-position tracking is achieved by T, the gain need only compensate for disagreement before T. Define

$$\gamma_{\text{req}} = \sup_{t_0 < t < T} \frac{\ln\left(\rho_0 / s(t)\right)}{t - t_0}. \tag{30}$$

Corollary 2 (Sufficient coupling selection)**.** *A sufficient condition for Theorem 3 is*

$$1 + \lambda_{\min}(M)\, \underline{w}\, \lambda_2(L_{\mathcal{H}}) > \gamma_{\text{req}}. \tag{31}$$

For $M = mI_d$, if $\gamma_{\text{req}} > 1$, it is sufficient to choose

$$m > \frac{\gamma_{\text{req}} - 1}{\underline{w}\lambda_2(L_{\mathcal{H}})}. \tag{32}$$

Thus, once the formation position plan and physical limits (d_{safe}, R_c) are specified, the margin and required rate can be computed offline and used to select a gain satisfying the certificate.

Proof. By Theorem 3, it suffices to ensure that $\rho_{\mathcal{H}}(t) < s(t)$ throughout the maneuver. By the definition of γ_{req} in (30), $\gamma_{\mathcal{H}} > \gamma_{\text{req}}$ ensures that

$$\rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)} < s(t), \qquad t \in (t_0, T).$$

Since $\rho_{\mathcal{H}}(t) \leq \rho_0 e^{-\gamma_{\mathcal{H}}(t-t_0)}$ by (26), the sufficient conditions hold on (t_0, T). The strict initial condition handles $t = t_0$. For $t \geq T$, exact formation-position tracking gives $\rho_{\mathcal{H}}(t) = 0 < s(t)$ by geometric feasibility. Thus, the conditions of Theorem 3 hold for all $t \geq t_0$. In particular, its minimum margin

$$\Delta_{\min} := \min_{t \in [t_0, T]} [s(t) - \rho_{\mathcal{H}}(t)] \tag{33}$$

is strictly positive, quantifying the available clearance of the certificate. □

Remark 2. *If $\rho_0 = 0$, geometric feasibility alone suffices. The gain condition is sufficient, not necessary; it cannot remedy a failed initial-margin test $\rho_0 < s(t_0)$.*

V. Numerical Validation

The results are evaluated in simulations of $n = 10$ agents in $\mathbb{R}^2$. For reporting, $u_i := \dot{x}_i$ denotes the commanded velocity supplied to a hypothetical low-level motion controller; $\|u(t)\|_2 = (\sum_i \|u_i(t)\|^2)^{1/2}$. The code integrates $J_u = \int_{t_0}^{t_f} \|u(t)\|_2^2 dt$ over the full simulation interval $[0, 12]$ s. Solver settings and benchmark parameters are in Table I. The required rates and certificate margins are numerical estimates from scalar optimization and sampled trajectory evaluation; they are not validated continuous-time global bounds. The complete simulation framework and experiments are available at `https://github.com/yash-purswani/finite-time-swarm-control`.

Two experiment categories are considered: (i) finite-time tracking and coupling-gain trade-offs and (ii) transient collision avoidance and connectivity preservation during aggressive star compression and expansion. Unless stated otherwise, initial perturbations satisfy $\rho_0 < s(0)$. The star formation rotates at rate ω and scales as $a(t) = a_\infty + (a_0 - a_\infty)e^{-\kappa t}$. Connectivity is assessed using a 15-edge connected spanning subgraph with $\lambda_2(L_{\mathcal{H}}) \approx 0.9316$, yielding the certified disagreement decay rate $\gamma_{\mathcal{H}} = 1 + 0.1540m$.

A. Finite-Time Tracking and Coupling-Gain Trade-offs

The influence of coupling gain m is evaluated during rotating star compression ($\kappa = 2.0$, $a_0 = 1.8$ to $a_\infty = 0.8$). As shown in Figure 1 and Table II, the tracking error satisfies the common theoretical deadline for all tested gains. The reported $t_{10^{-6}}$ is the first time at which $\|\delta(t)\|_2 \leq 10^{-6}$; it is a numerical tolerance metric, not verification of exact equality. Increasing m accelerates disagreement decay, while the theoretical deadline remains unchanged.

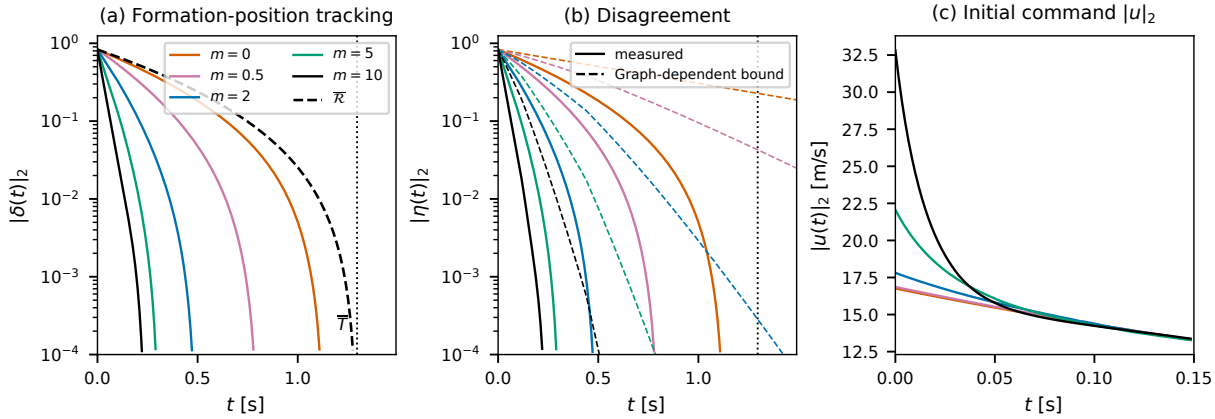


Fig. 1: Finite-time tracking, disagreement decay, and commanded-velocity magnitude across coupling gains m ($\kappa = 2.0$): (a) $\|\delta(t)\|_2$ and theoretical bound $\overline{\mathcal{R}}(t)$ with deadline T; (b) $\|\eta(t)\|_2$ and graph-dependent bounds; (c) aggregate commanded-velocity magnitude $\|u(t)\|_2$ over the initial 0.15 s.

TABLE I: Benchmark simulation parameters.

Param.	Value	Description
n	10	Swarm size in $\mathbb{R}^2$
β	0.5	Fractional-power velocity exponent
d_{safe}	1.0 m	Minimum pairwise collision threshold
R_c	6.0 m	Maximum communication radius
ε_w	0.05	Proximity weight regularizer
ν	1	Distance exponent in interaction weights
$\underline{w}$	0.1653 m^{-1}	Uniform lower bound $1/(R_c + \varepsilon_w)$
$r(t)$	$[t,\ 2\sin(0.5t)]^{\mathsf{T}}$	Reference centroid trajectory
ω	$\pi/4$ rad/s	Star template angular rotation rate
M	mI_2	Cooperative coupling matrix ($m \geq 0$)
$\mathcal{H}$	15 edges	Five-cycle on inner agents; each outer joins two adjacent inner agents
Solver	DOP853	`rtol` $= 10^{-6}$, `atol` $= 10^{-8}$, `max_step` $= 0.02$ s

TABLE II: Settling times and aggregate commanded-velocity metrics across coupling gains.

m	$t_{10^{-6}}$ (s)	T (s)	Peak $\|u\|_2$ [m/s]	J_u [m^2/s]
0.0	1.1517	1.3621	16.9828	429.44
0.5	0.8117	1.3621	17.0860	429.92
2.0	0.5250	1.3621	18.0322	431.55
5.0	0.4467	1.3621	22.2186	434.95
10.0	0.3933	1.3621	32.8156	440.74

B. Transient Safety and Connectivity

Rapid star compression and expansion are evaluated at $\kappa = 15.0$, with scales $a_0 = 1.80$ to $a_\infty = 0.8$ and $a_\infty = 3.00$, respectively. We compare certified gains ($m = 70$) with weak coupling ($m = 1$); the $m = 30$ cases are observed safe in both maneuvers despite failing the sufficient certificate. The reported $\Delta_{\min}$ values are numerical estimates evaluated over sampled trajectories.

VI. Conclusion

This paper developed a framework for finite-time tracking of arbitrary prescribed formation trajectories, with transient safety and connectivity certificates. Local feedback guarantees exact formation-position tracking within a graph-independent deadline, while cooperative coupling accelerates disagreement decay without altering this guarantee. The disagreement bound yields sufficient conditions for collision avoidance and preservation of a selected connected spanning subgraph, together with an offline coupling-gain selection rule based on planned geometric clearance and initial disagreement. Numerical experiments demonstrate tracking within the common settling-time bound across coupling gains and demonstrate faster convergence with increased control effort. Aggressive formation compression and expansion illustrate how appropriately selected coupling gains can certify collision avoidance and connectivity, while insufficient coupling may lead to constraint violations. The certificates are sufficient but not necessary, and their failure does not imply unsafe behaviour. Although the gain-selection rule provides a lower bound on coupling, actuator limits may impose practical upper bounds. Extending these guarantees to higher-order dynamics and actuator-constrained systems remains future work.

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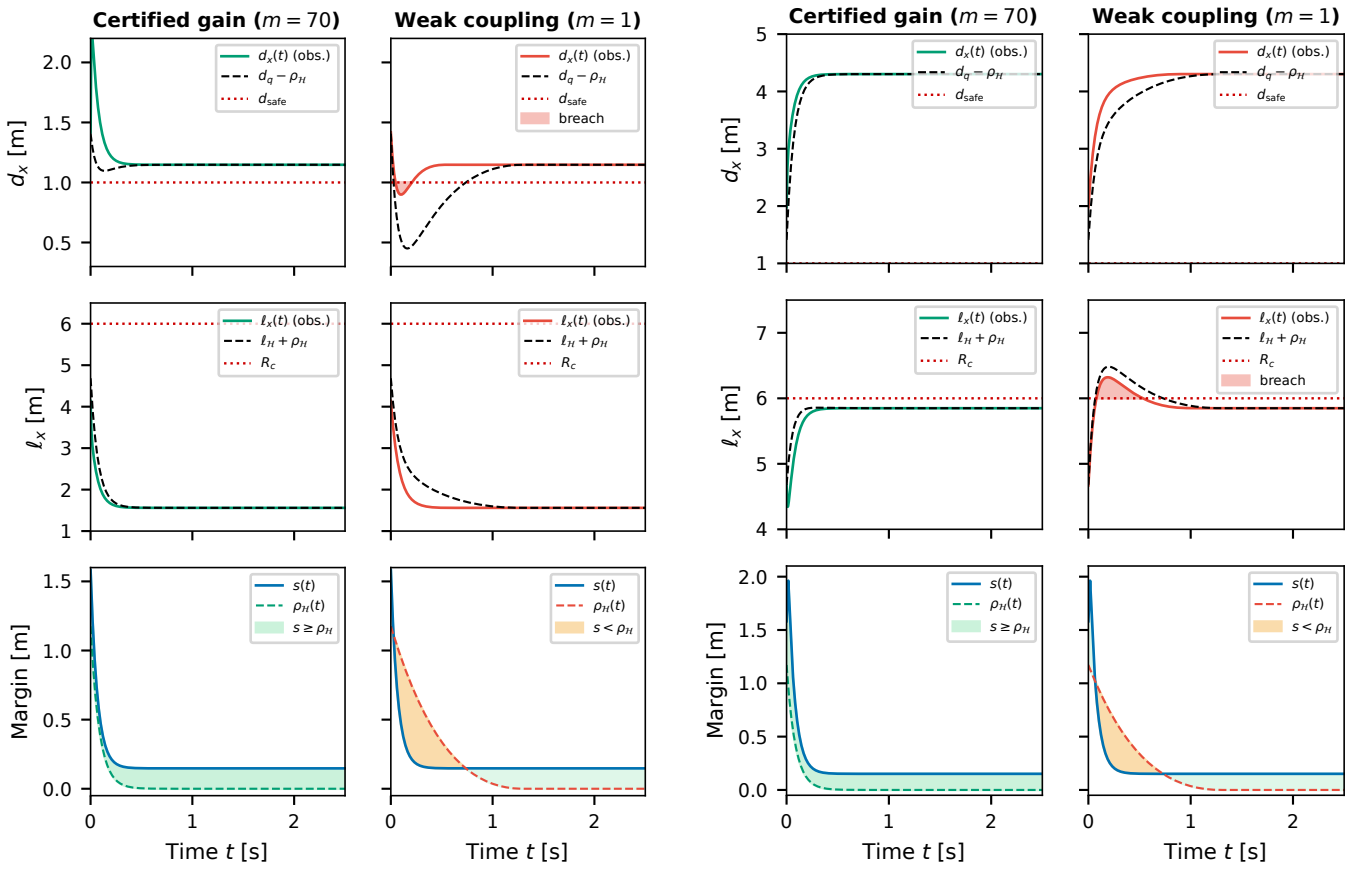
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(a) Formation compression ($a_0 = 1.8 \to a_\infty = 0.8$).

(b) Formation expansion ($a_0 = 1.8 \to a_\infty = 3.0$).

Fig. 2: Transient certificates during star compression and expansion ($\kappa = 15.0$). In each panel, columns show certified-gain and weak-coupling cases; rows show minimum pairwise distance, maximum selected-edge distance, and geometric margin versus disagreement bound. A collision is observed for weak coupling during compression, and a range violation during expansion.

TABLE III: Numerically evaluated certificates and observed geometry for aggressive star maneuvers ($\kappa = 15.0$).

Maneuver	Case	m	κ	$\gamma_{\mathcal{H}}$	γ_{req}	t_{10-6} (s)	$d_{x,\min}$ (m)	$\ell_{x,\max}$ (m)	$\Delta_{\min}$ (m)	Observed outcome
Compression	Certified gain	70	15	11.7786	9.3136	0.1750	1.1473	4.1387	**0.0970**	Certified; safe, connected
	Uncertified gain	30	15	5.6194	9.3136	0.1783	1.1473	4.1387	-0.2150	Observed safe, connected
	Weak coupling	1	15	1.1540	9.3136	0.5567	0.8990	4.1387	-0.5500	Collision observed
Expansion	Certified gain	70	15	11.7786	7.4220	0.2300	1.9785	5.8490	**0.1427**	Certified; safe, connected
	Uncertified gain	30	15	5.6194	7.4220	0.4000	1.9785	5.8490	-0.1180	Safe, Observed connected
	Weak coupling	1	15	1.1540	7.4220	1.0450	1.9785	6.3214	-0.4836	Range violation observed

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